AgentRoute

Learning-based semantic capability routing for AI agents in mobile networks

Ömer Faruk Gedikli · Bekir Emre Sarıpınar · Bekir Göktepe

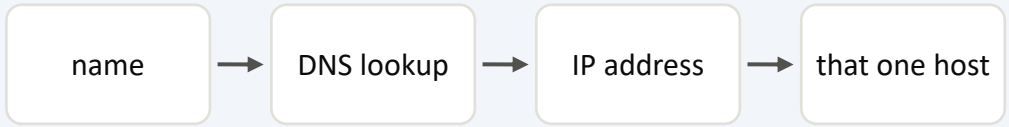
CSE 476/575 Mobile Communication Networks · 11 November 2026

The Internet asks where; agents ask who can

Today: routing to a location

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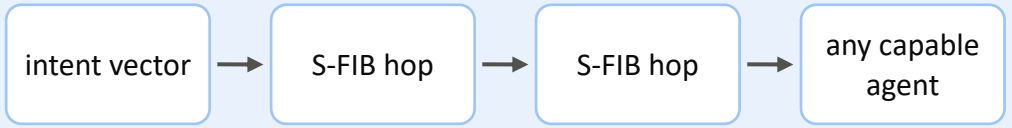
- A host has an IP address
- DNS maps names you already know to addresses
- Routers match address prefixes in their FIB
- The postman delivers to an address, not to someone who can help



Agents: routing to a capability

“find the cars in this photo”

- Who can do this task? Name and location unknown
- Matching by meaning, not by exact strings
- Machine speed, large and changing populations
- Many agents run on phones, cars and drones that change cell



Two classic answers break at scale

1 Central registry

Every registration and request passes one server: a bottleneck and a single point of failure.

202 ms

per request at 10,000 agents (AgentRoute: 21 ms)

2 Flooding

Ask every router within a TTL: works in small networks, explodes in large ones.

1,230

messages per request at 10,000 agents (AgentRoute: 315)

3 Mobility

Agents on phones, cars and drones keep changing cell, so any location table is soon stale.

47 / s

handovers with 10,000 agents, half of them mobile

AgentRoute: no central server, no flooding, and a handover only updates its neighbourhood

Related work stays above the network layer

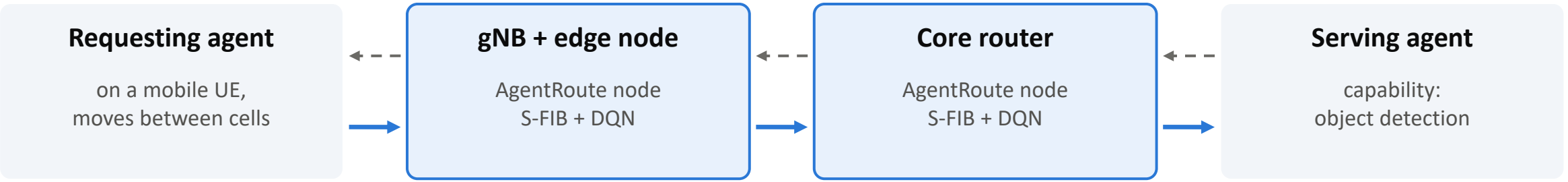
Work	Where it lives	Matching	Next hop	Mobile evaluation
[1] Layered protocol for the Internet of Agents (2025)	above HTTP	agent descriptions	–	–
[3] Agent discovery in the Internet of Agents (2025)	discovery service	registry / search	–	–
[4] IETF draft-wang-dmsc-drisac (2026)	registration service	semantic discovery	–	–
[2] SPFR: semantic potential field routing (2026)	network layer	semantic	heuristic	–
AgentRoute (ours)	**network layer**	**semantic S-FIB**	**learned (DQN, GNN)**	**cellular / edge, handovers**

References: [1] Fleming et al., arXiv:2511.19699 · [2] arXiv:2608.25396 · [3] Guo et al., arXiv:2511.19113 · [4] IETF draft, Aug. 2026

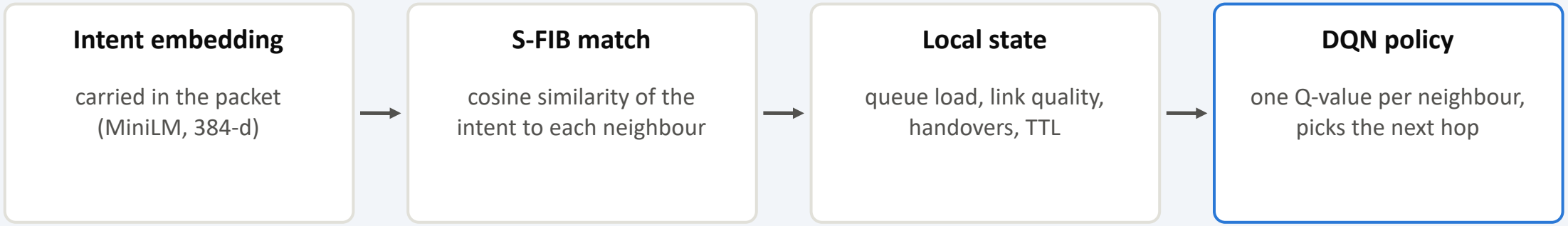
AgentRoute in one picture

(1) dashed: capability advertisements fill every S-FIB

(2) solid: a task packet with an intent embedding, hop by hop



Inside an AgentRoute node



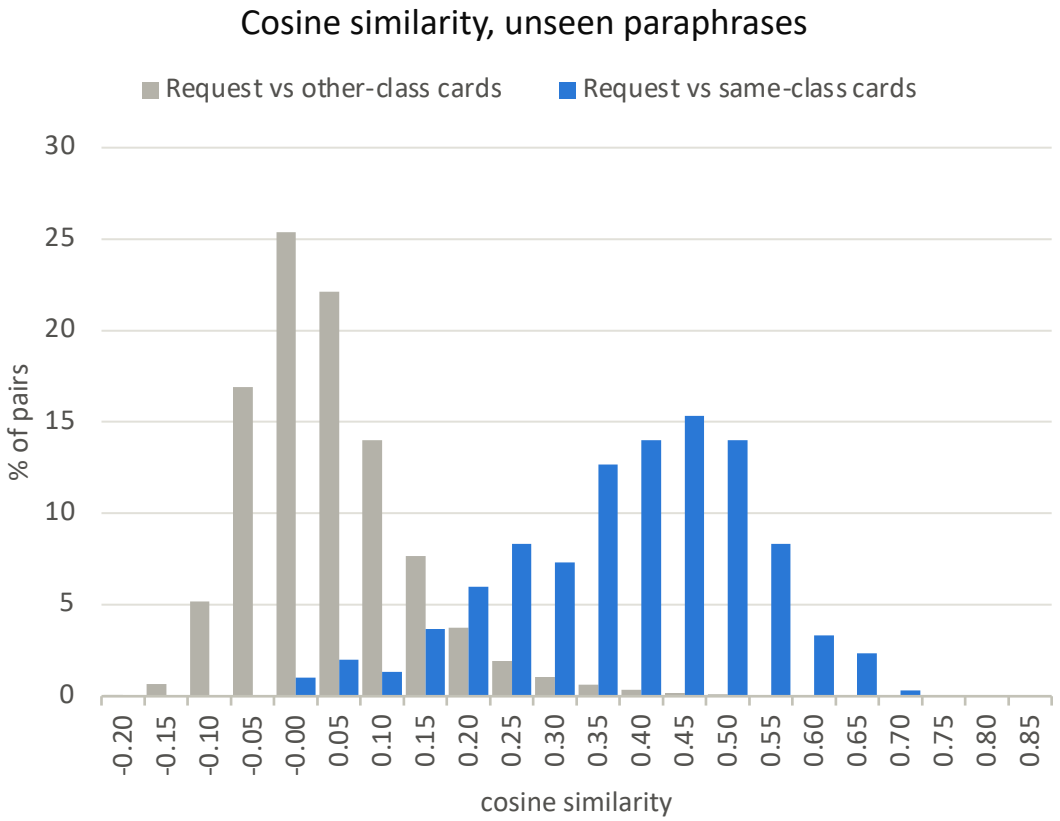
A capability address is a vector, not a host

Agent card → MiniLM → mean of the sentence vectors

- object detection in images
- box the vehicles in a traffic camera image
- show where each person stands in a crowd photo

Request “find the cars in this photo” (accept if cos ≥ τ = 0.405)

object detection in images	**0.517 ✓ accept**
locate and label every object in a photo with bounding boxes	**0.409 ✓ accept**
image classification into categories	**0.393 ✗ reject**



τ = 0.405 · top-1 class accuracy 77% · false accept 0.6%

The S-FIB: what lies behind each neighbour

Neighbour	k = 4 representative capabilities	Min hops
gNB 7 (Xn)	object detection · OCR · face recognition · image captioning	1
gNB 9 (Xn)	translation · summarization · speech-to-text · grammar	2
Core 12 (backhaul)	cluster centres of everything within R hops, incl. the cloud	2

Illustrative S-FIB of one gNB + edge router

49%

greedy success with a mean summary

51%

with k = 4 representatives

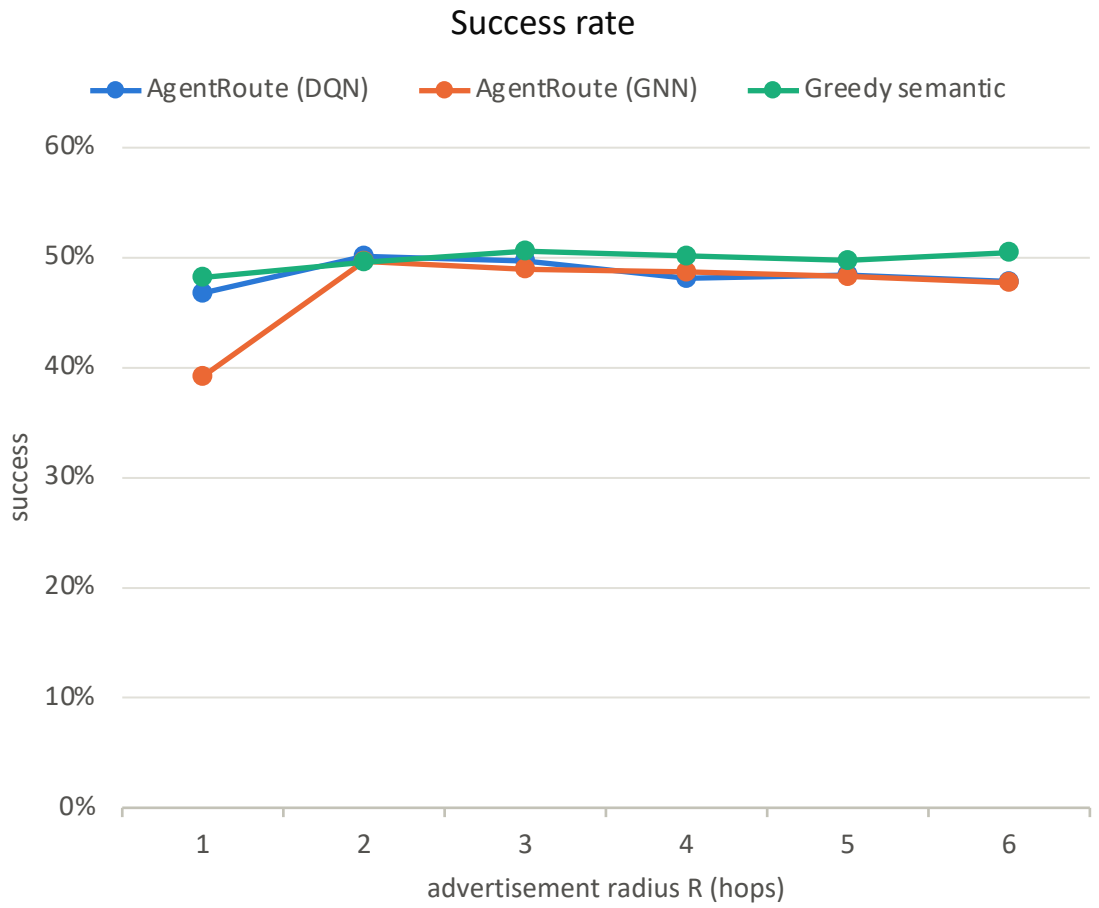
56%

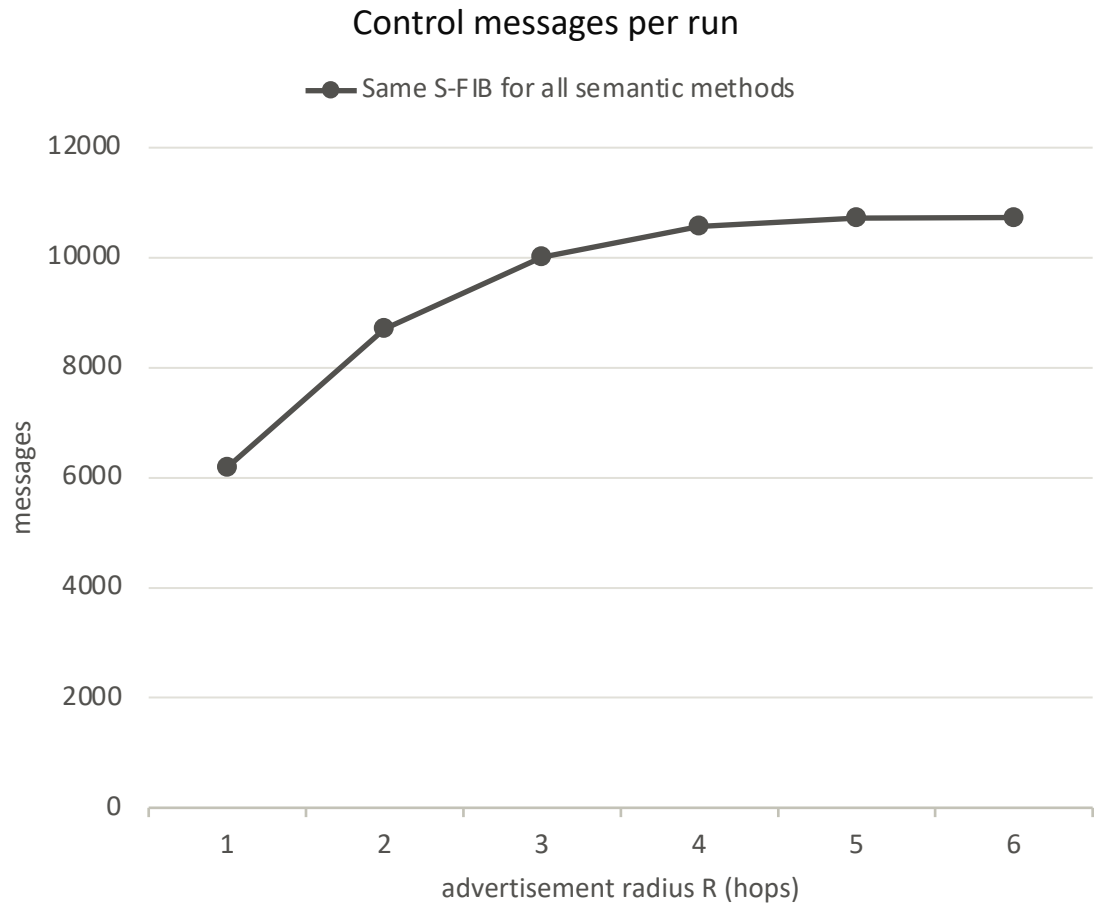
with the full table

IP FIB vs S-FIB

- IP: prefix → output port
- S-FIB: neighbour → capability summary
- Score of a neighbour = best cosine with its representatives
- k-means keeps distinct capabilities apart; a mean blurs them
- Fixed size per neighbour, independent of the agent count

Advertisements reach R hops, not the whole network





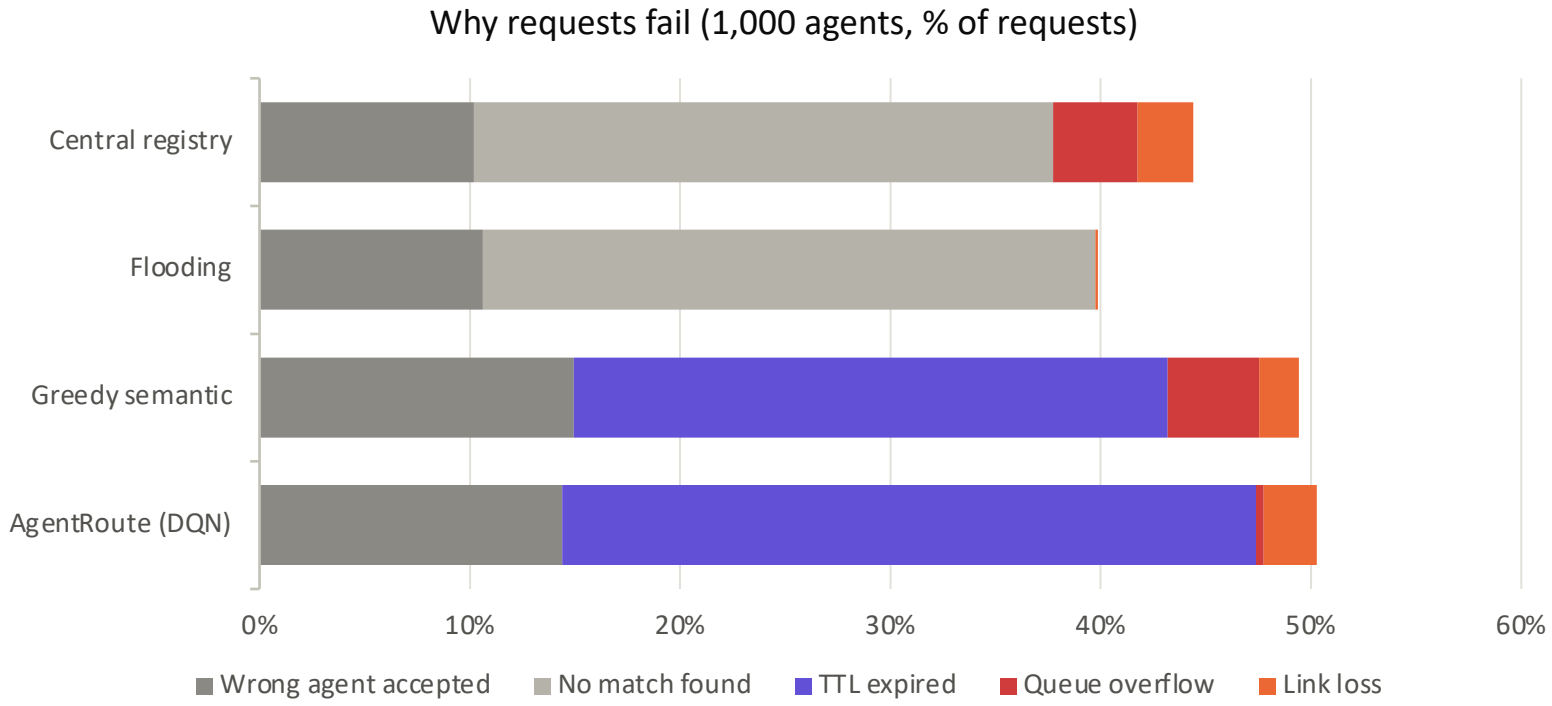
Greedy works until the best neighbour is busy

51%

greedy success with 1,000 agents

56%

central registry, same runs

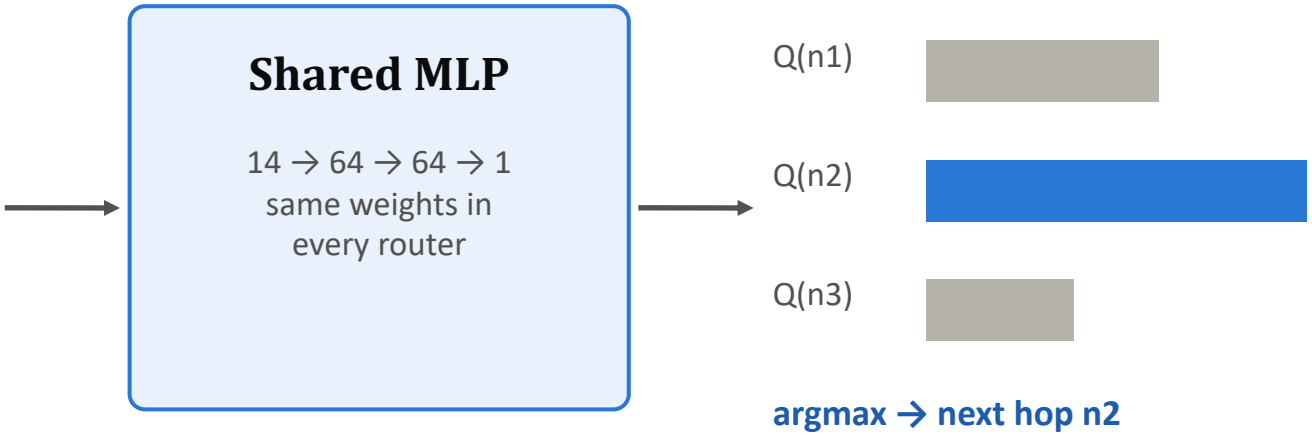


What if the most similar neighbour is congested, or the device is in the middle of a handover?

One shared DQN scores every neighbour

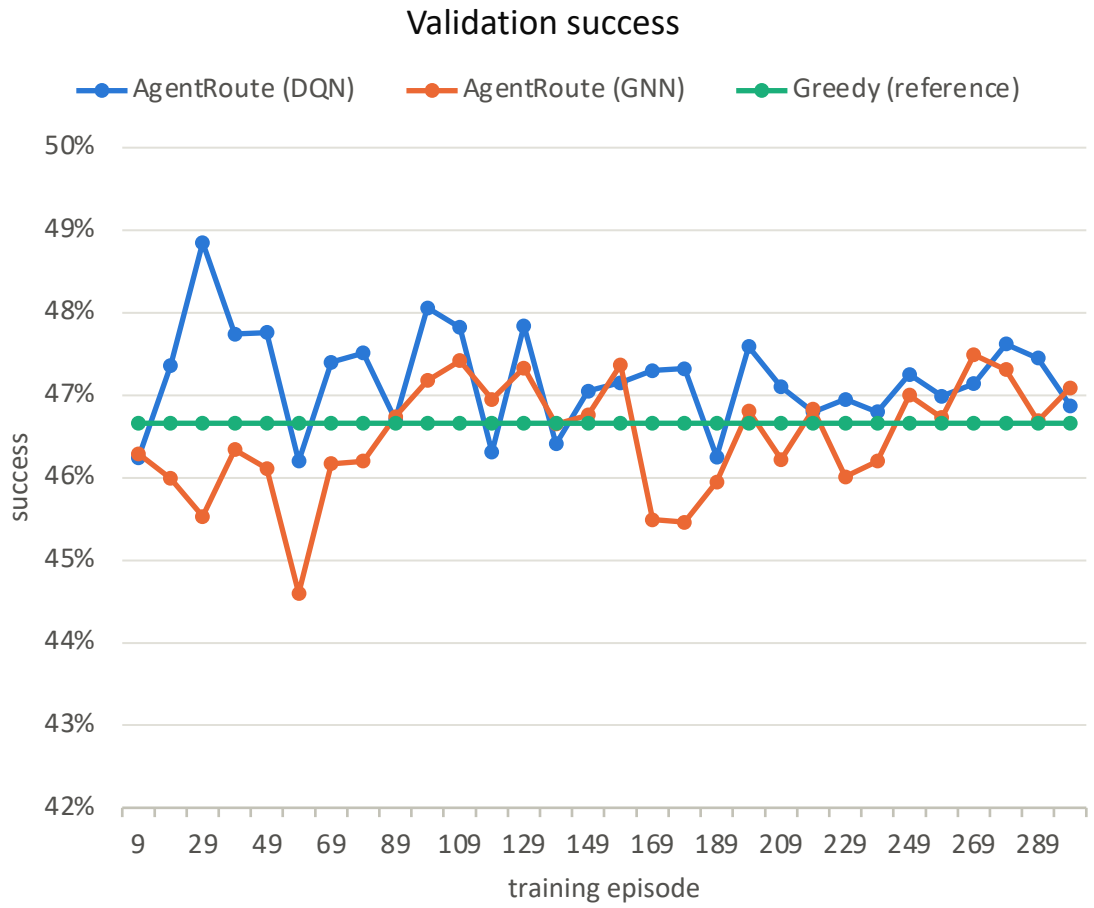
Features of one neighbour (14)

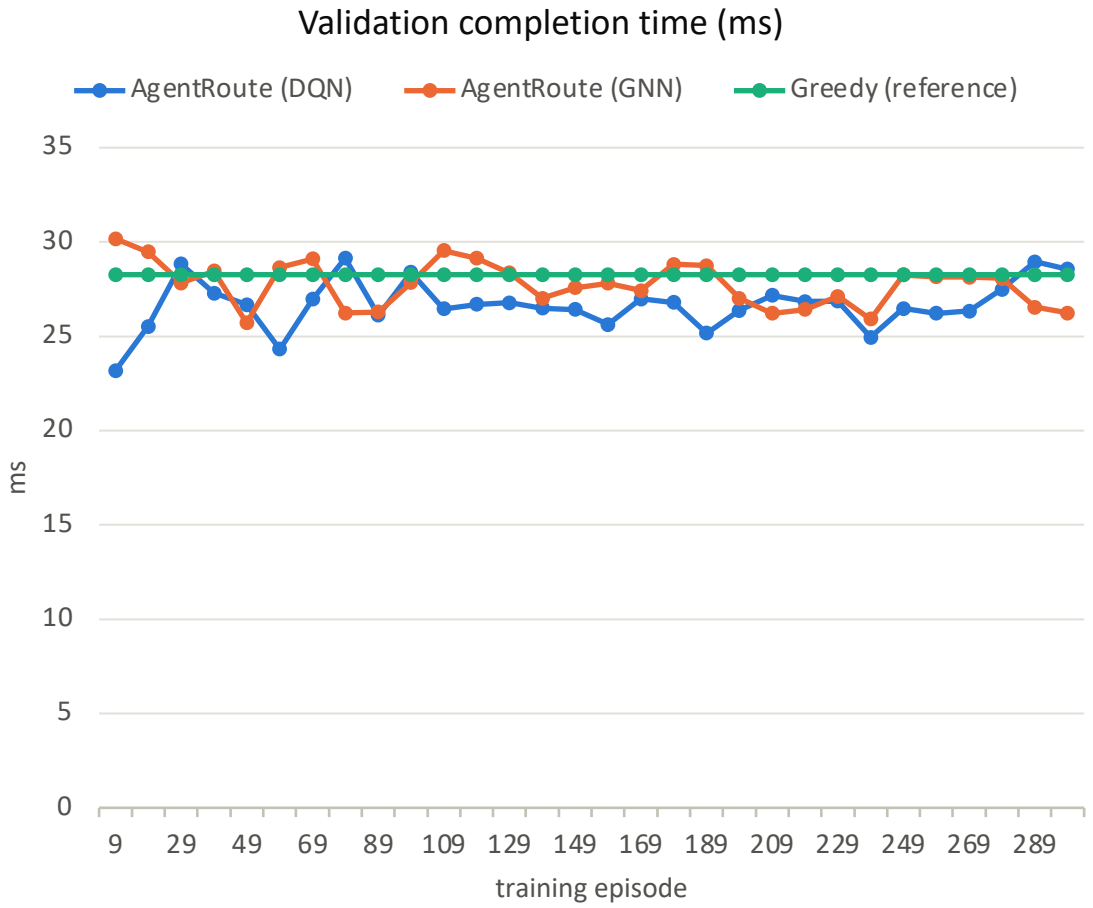
- S-FIB similarity: max, mean, gap to best
- match above τ, hop distance
- queue occupancy (hello)
- link quality and delay
- neighbour in handover
- TTL left, already visited
- node type: gNB / core / cloud



Reward per hop −(0.05 + delay / 100 ms) · delivered +1 · dropped −1
Training Double DQN, replay 50k, target net, ε 1 → 0.05, 300 random scenarios

Training: the DQN matches greedy and delivers faster





A handover only updates its neighbourhood

Incremental S-FIB update

1. Old cell withdraws the agent within R hops
2. New cell announces it within R hops
3. Messages only where a summary really changes

24.4
messages per handover at 10,000 agents

0.8%
of a full re-advertisement (2,892 messages)



Evaluation setup

Topology

gNB + edge cells, core grid, cloud; 9 → 400 cells, up to 450 routers; queues, bursty hotspots, lossy links

Agents

40% fixed edge, 50% mobile (random waypoint, handover with hysteresis), 10% cloud; Zipf capability supply

Requests

150 unseen paraphrases of 50 capability classes, 10 requests/s, TTL 16

Baselines

central registry (DNS-like), TTL-limited flooding, Chord exact match, greedy semantic (same S-FIB)

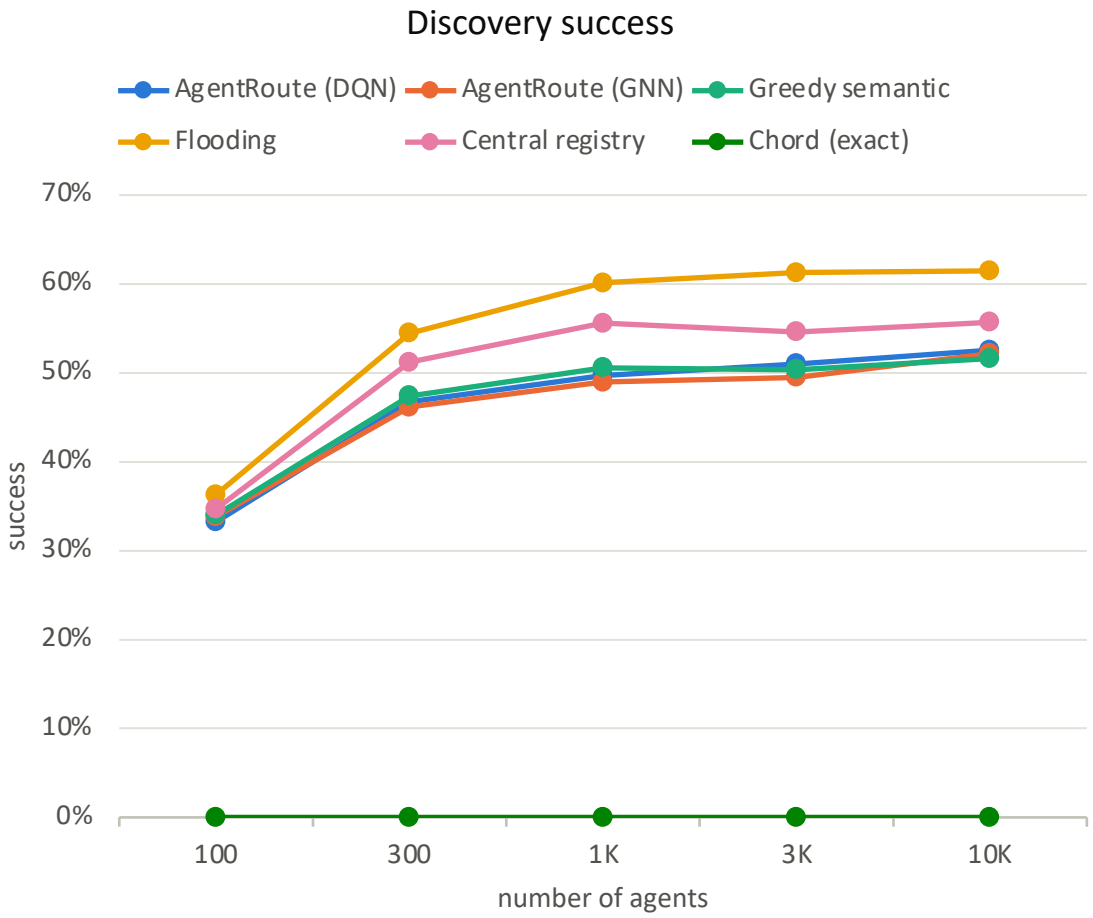
Metrics

discovery success, completion time, hops, control + query messages

Runs

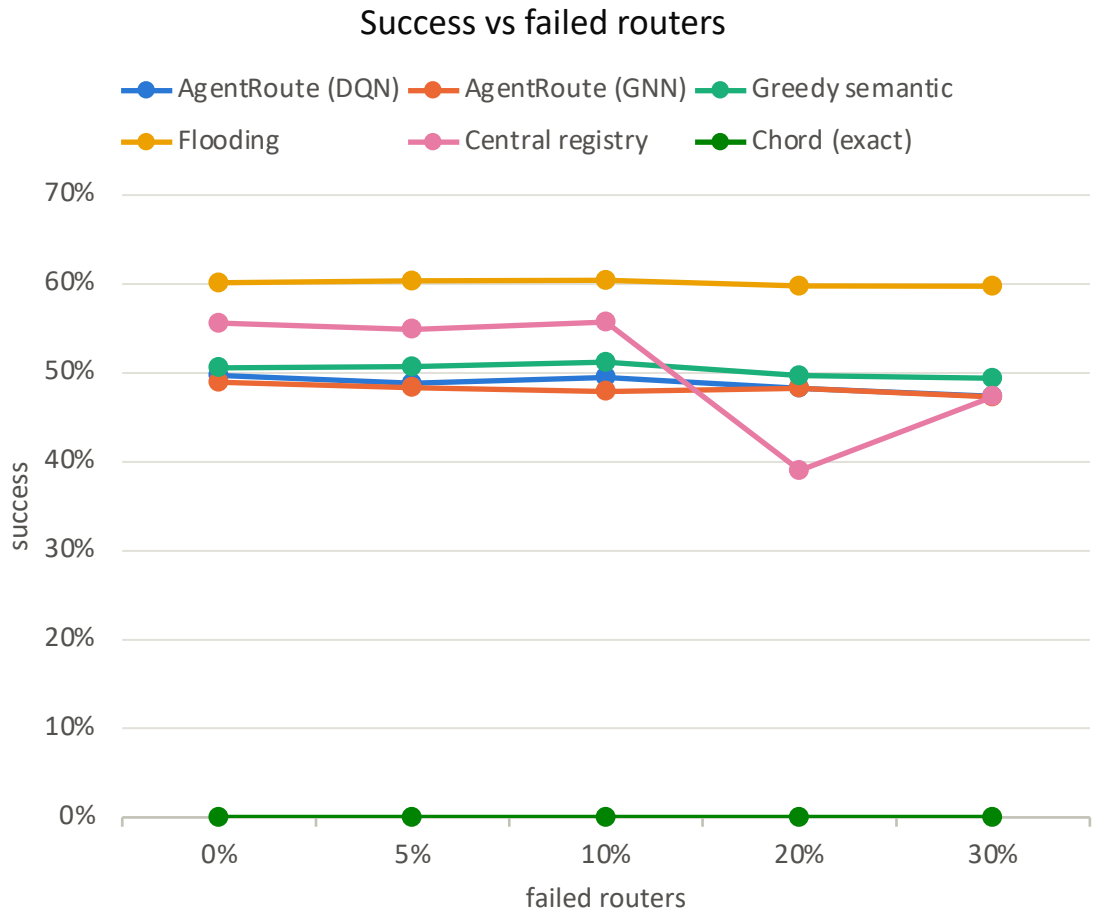
SimPy + NetworkX, 5 seeds per point, 95% CI, 100 → 10,000 agents, laptop CPU

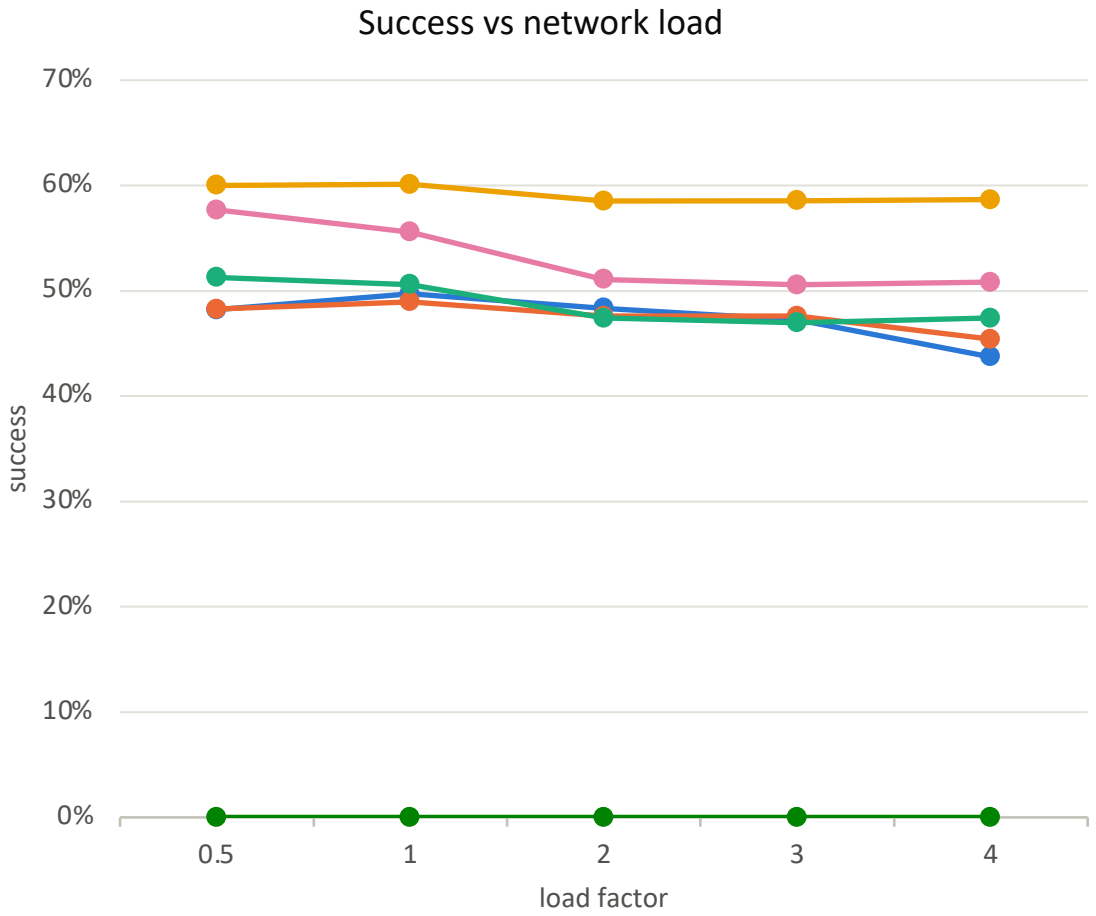
Scaling to 10,000 agents



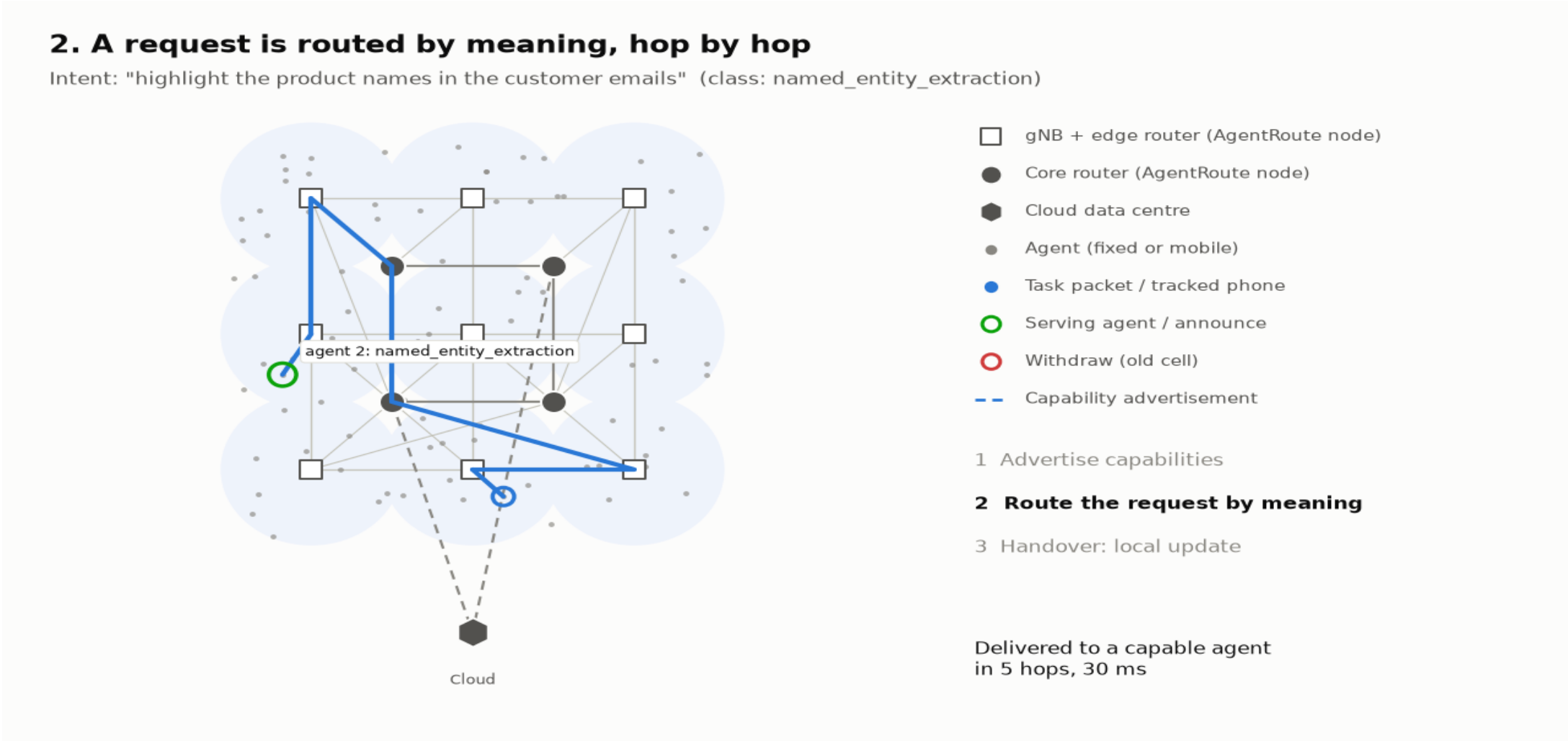


Robustness: failures and load

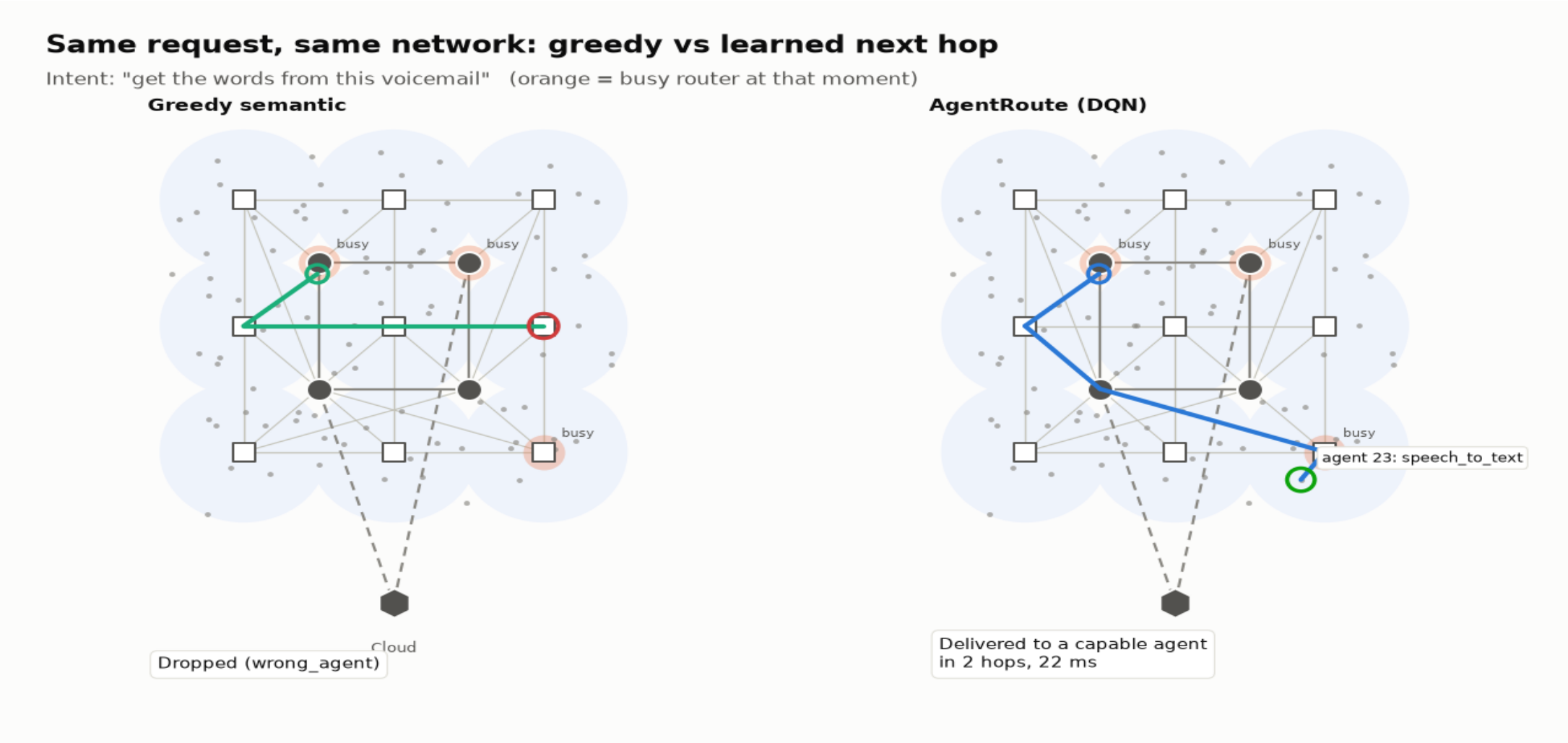




Demo: a request routed by meaning



Same request: greedy vs learned next hop



Takeaways

1. Capability addresses + S-FIB find agents by meaning: 53% at 10,000 agents, where exact-match Chord finds 0%

2. No central server and no flood: 315 messages per request vs 1,230 for flooding; 21 ms vs 202 ms for the registry

3. One shared DQN learns the next hop: as successful as greedy and 7–14% faster, but it trails greedy under the heaviest load

Future work: on-device fine-tuning · signed capability advertisements · UDP overlay prototype · better encoders

Backup: S-FIB summary and header size

